Lasers in Medical Science (2026) 41:68
https://doi.org/10.1007/s10103-026-04842-3

REVIEW

Effectiveness of high-intensity laser therapy for tendinopathy: a systematic review and meta-analysis of randomised controlled trials

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Received: 15 September 2025 / Accepted: 19 February 2026
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Abstract

Tendinopathy is a common musculoskeletal disorder causing significant pain and functional impairment. High-intensity laser therapy (HILT) has increasingly been employed for tendinopathy. However, further research is required to elucidate therapeutic efficacy differences across various subtypes, determine optimal treatment parameters, and assess long-term effects. To evaluate the effectiveness of HILT for tendinopathy. PubMed, Embase, Cochrane Library, China Biology Medicine, Web of Science, CNKI, and SCOPUS databases were searched to include randomised controlled trials comparing HILT with placebo or other treatments up to 20 October 2025. Primary outcome measures included pain intensity and disability, analyzed as MD and SMD with 95% confidence intervals, respectively. Risk of bias was assessed using the Cochrane Tool and Physiotherapy Evidence Database scale. Heterogeneity was assessed with I^2, and random-effects models were used. 15 studies involving 629 patients were included: lateral epicondylitis (53%), rotator cuff tendinopathy (27%), and others (20%). Studies showed moderate quality (mean PEDro score 6.3), with strong randomisation (100% low risk) but weaknesses in allocation concealment (20% low risk) and blinding. HILT significantly improved pain (MD: -1.15; 95%CI: -1.73 to -0.58) and disability (SMD: -1.00; 95%CI: -1.77 to -0.22, $p<0.05$). Pain reduction exceeded the MCID, though the lower confidence interval bound suggests marginal benefit in some. For disability, subgroup analysis by anatomical site revealed greater improvement in shoulder-related conditions (MD: -2.54) than in elbow-related conditions (MD: -0.91). Stress pain showed the most substantial relief (MD: -2.30), and treatment effects increased over time, peaking after 16 weeks. HILT effectively improves pain and disability in tendinopathy, with site-specific and cumulative benefits. Given methodological limitations, future high-quality studies are needed to optimize parameters and confirm long-term benefits.

Keywords high-intensity laser therapy · tendinopathy · pain · systematic review · meta-analysis

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PROSPERO registration and date: CRD420251135664 on 25 August 2025.

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Published online: 11 April 2026

Introduction

Musculoskeletal pain ranks among the primary causes of disability and diminished quality of life worldwide. Among numerous musculoskeletal conditions, tendinopathies are particularly prevalent, with their global incidence showing an upward trend. According to epidemiologic surveys, approximately 15–20% of adults have suffered from tendinopathy [1]. Especially in the athletic population, this percentage can be as high as 30–50% [2]. Tendinopathy not only affects the patient's daily life and ability to work, but can also lead to chronic pain and dysfunction. Studies have shown that patients with tendinopathies have a significantly reduced quality of life, with approximately 40% of them reporting reduced work productivity and 25% interrupting their physical activity due to pain [3]. Economic analyses have shown that tendon-related disorders are responsible for more than $8.7 billion in healthcare expenditures and lost productivity globally each year [4]. The socioeconomic burden of tendinopathies is expected to increase further as the population ages and sedentary lifestyles become more prevalent. In addition, the poor blood supply and low cell density of tendon tissues result in their limited self-healing ability, predisposing tendinopathies to chronicity and recurrence [5].

The current treatment of tendinopathy is varied and includes conservative treatment and surgical intervention. Conservative treatments include rest, physical therapy, nonsteroidal anti-inflammatory drugs, corticosteroid injections, and extracorporeal shock wave therapy [6]. However, these traditional treatments have obvious limitations: medication may bring gastrointestinal discomfort, renal impairment and other side effects; corticosteroid injections can provide short-term relief, but the long-term effect is poor, and there is a risk of tendon rupture [7]; and the effectiveness of extracorporeal shockwave therapy has individual differences, and the process of treatment may be accompanied by pain [8]. Surgery may be an option for refractory cases, but it is traumatic, has a long recovery period, and carries a high risk of complications [9].

In recent years, HILT has attracted widespread attention as an emerging non-invasive treatment method [10–12]. HILT typically employs near-infrared wavelengths (e.g., 808–1064 nm), which provide greater tissue penetration than the wavelengths commonly used in low-level laser therapy. Combined with higher power output, this enables more pronounced photothermal and photobiomodulatory effects at depth, promoting tissue repair and reducing inflammation and pain [13]. Studies have shown that high-intensity lasers can accelerate the tendon healing process by promoting capillary formation through photobiomodulation, increasing the level of oxyhemoglobin, and stimulating the proliferation of fibroblasts [14, 15]. At the same time, its thermal effect can improve local blood circulation, reduce muscle spasm, and improve the ductility of collagen fibres [16, 17]. Current clinical research evidence regarding HILT for tendinopathy remains limited, with considerable variation in outcomes across studies. This research aims to provide an updated synthesis of evidence for HILT treatment of tendinopathy through systematic review and meta-analysis. Compared with previous reviews, this study uniquely contributes by undertaking the first multidimensional, in-depth exploration: through subgroup analysis, it systematically compared the efficacy differences of HILT across various anatomical sites; clarified its differential therapeutic effects on distinct pain states; and revealed the cumulative effect patterns of its therapeutic outcomes over time. These findings not only confirm the overall efficacy of HILT but, more importantly, provide crucial evidence for formulating individualised, precision treatment plans in clinical practice.

Methods

This study constitutes a systematic review and meta-analysis, with the detailed agreement registered in PROSPERO on 25 August 2025. The registration numbers are CRD420251135664. This systematic review was carried out according to PRISMA guidelines (Supplementary Material 3).

Literature search

Literature was retrieved from PubMed, Embase, Cochrane Library, China Biology Medicine, Web of Science, and China Knowledge Network (CNKI) and SCOPUS up to 20 October 2025. The MeSH terms and free text words included in the search strategy are shown in Supplementary Material 1. To maximise the sensitivity of the included articles, only intervention-related terms were used for article retrieval. Search terms included "high-intensity laser therapy", "high-energy laser", and "high-power laser". The terms "tendinitis", "tendon injury", "tennis elbow", "rotator cuff injury", "tenosynovitis", "achilles tendonitis", "tendinopathy", "tendon attachment point inflammation", "patellar tendinopathy", and "golfer's elbow" were also applied to the search strategy when searching for tendinopathy.

Study selection

The inclusion criteria were as follows: patients with a diagnosis of tendinopathy were included regardless of country, region, or ethnicity, with no language restrictions on published material. The diagnosis of tendinopathy specifically

included, but was not limited to, common conditions. This encompassed individuals with subacromial impingement syndrome confirmed by imaging, presenting with degenerative changes or partial-thickness tears of the supraspinatus tendon. All studies included in this review were randomised controlled trials in which the control groups received interventions such as placebo, alternative treatments (including extracorporeal shock wave therapy (ESWT), Low-Intensity Laser Therapy (LILT), kinesiologic taping, ultrasound (US)), drug injection or conventional physiotherapy.

Exclusion criteria were as follows: studies of whiplash syndrome diagnosed solely on clinical signs without imaging confirmation; pain scores were not reported; not a randomised controlled trial; reviews, conference abstracts, systematic reviews of animal studies and no outcome data, and repetition of unavailable data.

Data extraction

Two researchers independently entered data from the selected literature into a pre-designed data extraction form and cross-checked and integrated all information. Information on adverse events, including their type, frequency and attribution, was systematically collected. In case of disagreement, consensus was reached through negotiation or, if necessary, by inviting a third-party researcher to arbitrate. The extracted information mainly included: first author's name, year of publication, country, study sample size, patient characteristics, treatment modality, duration of follow-up, and outcomes.

Quality assessment

Two reviewers (Yi-Shan Xue and Yu-Han Liu) used the Cochrane Collaboration's Risk of Bias (RoB) tool and PEDro scale to assess the quality of RCTs. Any discrepancies were resolved through consensus or by consulting a third reviewer (Ying Li). The Cochrane RoB tool [18] was applied according to its standard guidelines, evaluating five domains. "low risk" was assigned only if all domains were judged as low risk. "high risk" was assigned if any domain was judged as high risk. In all other cases, the overall judgement was "some concerns". If the PEDro total scores for these trials are ≥ 7, 5–6, or ≤ 4, respectively, they are labelled as having "high", "moderate", or "poor" methodological quality [19].

Statistical analysis

In this study, the statistical analysis of the overall efficacy and subgroup effects of HILT intervention for tendinopathy was implemented using Review Manager 5.4. The data used for Meta-analysis were obtained directly from the included literature or based on the original data converted by the appropriate tools. The extracted means, standard deviations, and sample sizes were entered into the analysis software. All meta-analyses employed random-effects models, applying relative weighting to individual trial results even in the presence of statistical heterogeneity. Mean difference (MD) was used to indicate the effect size if the studies used the same instrument to measure the same outcome. For instance, pain scores are reported via the Visual Analogue Scale (VAS) and the Numerical Pain Rating Scale (NPRS). If the measurement instruments were different, the standardised mean difference (SMD) was chosen to combine the results of continuous variables. For example, disability outcomes reported by patients may be measured using disparate instruments. For studies with multiple intervention arms, control group data were split to avoid double-counting in pairwise comparisons, following Cochrane guidelines. Inter-study heterogeneity was assessed using the I^2 statistic [20]. For results with high heterogeneity, the source of heterogeneity was explored by sensitivity analysis or subgroup analysis. In order to identify publication biases, contour funnel plots were used.

Results

Literature search

From the initial literature search, a total of 308 articles were retrieved, and after removing duplicates, there were 264 articles, of which 105 did not match the study, 17 were animal trials, and 99 were previous reviews and meta-analyses, yielding 38 articles that matched the study. After full-text review of the 38 articles, 5 were not able to obtain full-text literature, 12 lacked outcome data, and 6 studies did not match the methodology. Ultimately, 15 studies were included in this systematic review, 13 for meta-analysis [21–33] and 2 for descriptive analysis [34, 35]. The specific screening and inclusion process is shown in Fig. 1.

General characteristics of included studies

The included studies comprised 1 study on Achilles tendinopathy [34], De Quervain tenosynovitis [31], and proximal hamstring tendinopathy (6.7% each) [29]. 4 on rotator cuff tendinopathy (26.7%) [22, 25, 33, 35], 8 on lateral epicondylitis (53.3%) [21, 23, 24, 26–28, 30, 33] (Table 1). All studies were randomised controlled trials. 1 study compared HILT with a placebo [24], 9 compared it with other

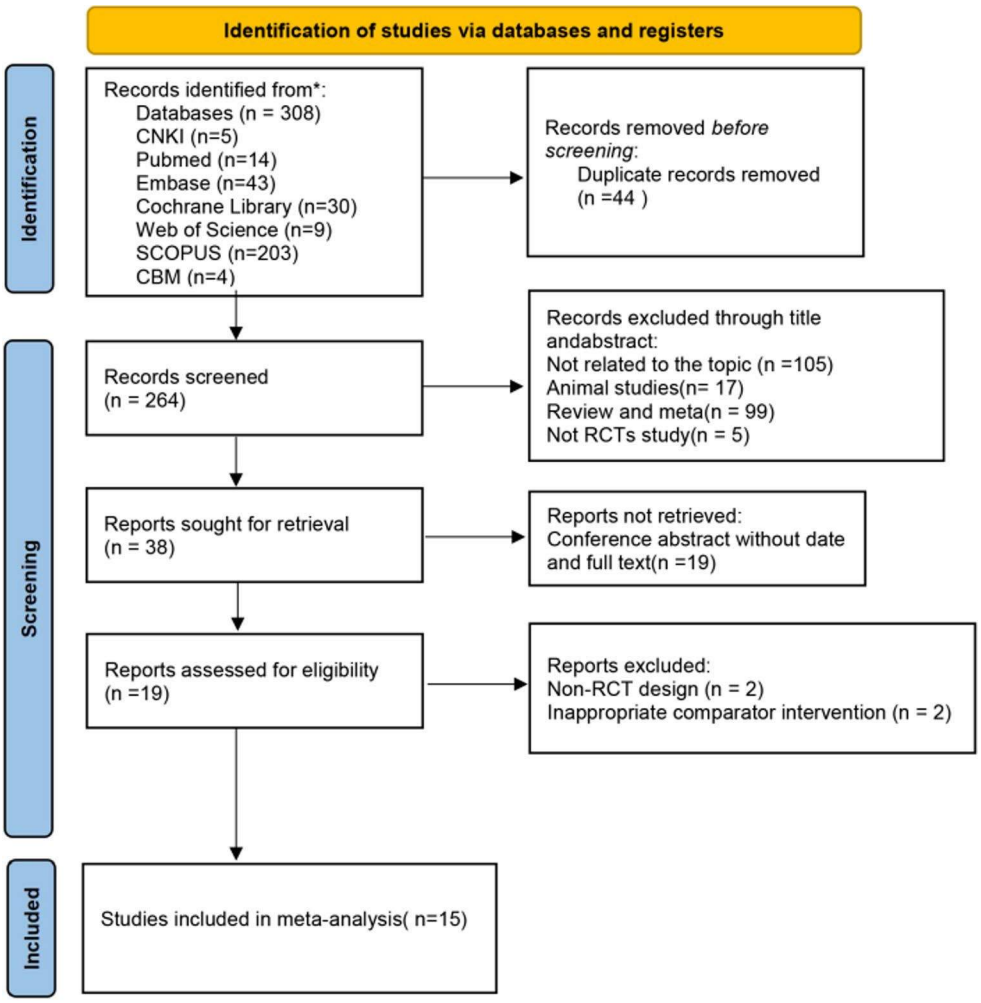


Fig. 1 Screening flowchart

treatment modalities [21, 23, 26–28, 30–33], and 3 compared HILT as an adjunctive intervention to conventional physiotherapy [22, 25, 29]. Sample sizes ranged from 14 to 35 participants across all studies, published between 2014 and 2025. Trials were conducted in Iran [22, 31, 35], Saudi Arabia [25, 29], Turkey [23, 24, 26–28, 30, 32], Italy [33, 34], China [21], and India [29].

Pain was assessed using the VAS [21–28, 30–35] and NPRS [29]. The 10-centimetre VAS measured pain on a scale ranging from 0 (no pain) to 10 (worst possible pain). Thus, higher VAS scores indicated greater pain severity. Functional activity was evaluated according to the treatment region. For rotator cuff tendinopathy, the Shoulder Pain and Disability Index (SPADI) [36] and Constant Murley Scale (CMS) [37] were employed; for lateral epicondylitis, the Patient-Related Tennis Elbow Evaluation (PRTEE) [38] and the Disability of the Arm and Shoulder (DASH) [39] questionnaire served as measures of functional impairment. The characteristics included in the study for HILT are summarised in Table 2.

Risk of bias assessment and quality

The risk of bias assessment revealed methodological quality variations across included studies. As shown in Fig. 2, randomised outcome generation demonstrated low risk of bias across all studies. 3 studies [24, 28, 33] reported allocation concealment (low risk of bias), whilst 10 studies failed to mention allocation concealment (risk of bias unclear). Regarding outcome assessment blinding, 6 retrieved studies [23–25, 29, 30, 33] were at low risk, 1 study was high risk [26], while 6 studies [21, 22, 28, 29, 31, 32] did not report (risk of bias unclear). Most studies had complete information on outcomes (low risk of bias), though 5 [23, 24, 27–29] reported patients who withdrew but were not included in the analysis (high risk of bias). No studies exhibited selective reporting bias or other biases. Among the RoB domains, random sequence generation was the best-rated (100% low risk), in contrast to allocation concealment, which was the worst-rated (20% low risk). Detailed PEDro quality assessment results are presented in Table 2 in the supplementary materials.

Table 1 Characteristics of included studies

Study	Country	Patients	intervention（Experimental group/Contril group）			Measures
Sai-Nan Zhang et al (2024) [21]	China	*n*=60Exp: 30（m16/f14)Con: 30（m17/f13)population：general population	HILT(Light Force, 810nm)1 time in a day with a total of 8 sessions	Infrared therapy device (XD-3000B)1 time in a day with a total of 8 sessions	——	Pain:VASDisability:-follow-up:3 mouths
Morteza Beyrami et al (2023) [22]	Iran	*n*=32 age:40-60Exp: 16（7m/9f)Con: 16（10m/6f)population：general population	PT+HPLT（GaAlAs,808nm）3 days in a week with a total of 12 sessions	PT	——	Pain:VASDisability:SPADIfollow-up:-
Emine Esra Bilir et al (2024) [23]	Turkey	*n*=40 age＞18Exp: 20（9m/11f)Con: 20（9m/11f)population：general population	HIRO TT (ASA,Italy)3 days in a week with a total of 9 sessions	ESWT1 days in a week with a total of 3 sessions	——	Pain:VAS(at rest,at motion,at nigh) Disability:QDASHfollow-up:6 weeks and 1 mouth
Umit Dundar et al (2015) [24]	Turkey	*n*=91 age:20-50Exp1: 30（17m/13f) Exp2: 31（17m/14f)Exp3: 30（15m/15f)population：general population	HILT（ASA Laser,1064nm）a day for 15 days during a period of 3 weeks	placebo HILT	the lateral counter-force brace for LE (Aurafix, Turkey) during the daytime for 4 weeks	Pain:VAS(at rest,under strain) Disability:PRTEEQfollow-up:12 weeks
A.M.Elsodany et al (2018) [25]	Egypt and Saudi Arabia	*n*=60 age:50.2±3.6 population：general population	YAG (ASA，1064nm) + ”sham” laser + exercise program3 sessions each week for 4 weeks with a total of 12 sessions	sham laser + exercise program3 sessions per week for 4 weeks	——	Pain:VASDisability:SPADIfollow-up:3 and 6 mouths
İbrahim Karaca et al (2022) [26]	Turkey	*n*=42 age：21-63Con: 14（8m/6f) Exp 1: 14 (7m/7f)Exp 2: 14（5m/9f) population：general population	PT+HILT(BTL-6000,UK)a total of 4 sessions for 2 week	PT+ESWT(BTL-5000 SWT POWER, UK)twice a week for a total of 4 sessions	PT	Pain:VAS(at rest,at motion) Disability:PRTEE-Tfollow-up:6 weeks
Angela Notar-nicolaet al (2014) [34]	Italy	*n*=60age:18-80Exp: 30（14m/16f)Con: 30（13m/17f) population：general population	CHELT (YAG, 1084, 810, 980 nm) 10 daily sessions	ESWT3sessions at 3- to 4-day intervals of 1,600 impulses with an energy flux density (EFD) of 0.05–0.07 mJ/mm2	——	Pain:VASDisability:-follow-up:2 and 6 mouths
Sidika Buyukvural Sen et al (2024) [28]	Turkey	*n*=50 age≥18Exp: 25（12m/13f)Con: 25（9m/16f)population：general population	HILT (YAG Laser,1064 nm)15 con-secutive sessions over the course of 3 weeks	ESWT(Roland series E-SWT 2)1 time in a week with a total of 3 sessions	——	Pain:VAS(at rest,at motion) Disability:Q-DASHfollow-up:12 weeks
Sachin Verma et al (2022) [29]	India and Saudi Arabia	*n*=36age:18-35population：athlete	HPLT (LiteCure, USA,980/810 nm) 3 days a week for a total of 3 weeks	PT:3 days a week fora total of 3 weeks	——	Pain:NPRSDisability:-follow-up:-
Ali SALLIet al (2016) [27]	Turkey	*n*=65 age:46.5±8.1Exp: 31（8m/23f)Con: 34（10m/24f) population：general population	HILT5 days in a week with a total of 10 sessions	Bangagea total of 6 weeks	——	Pain:VASDisability:QDASHfollow-up:-

Table 1 (continued)

Study	Country	Patients	intervention（Experimental group/Contril group）		Measures
H. Yüksel et al(2025) [30]	Turkey	*n*=55 age:48.1Exp1: 20（6m/14f) Exp2: 18（7m/11f)Con: 17（7m/10f) population：general population	HILT + TE: (Mectronic, Italy) three times a week for 3 weeks	KT + TE:twotimes a week for 3 weeks.	TEtwice a day for 6 weeks	Pain:VAS(at rest,at motion) Disability:Q-DASHfollow-up:4 weeks
Z. Zaki et al (2025) [35]	Iran	*n*=36 age:56.33±8.61Exp: 18（9m/9f)Con: 18（9m/9f) population：general population	HPLT(Cube 4 plus med, 610nm,800nm,905nm,970nm)+Exs	Sham + Exs	——	Pain:VAS(at rest,at motion,at nigh) Disability:SPADIfollow-up:-
M. R. Nikoo et al (2024) [31]	Iran	*n*=30 age:18-65Exp: 15（1m/14f)Con: 15（3m/12f) population：general population	HILT(GaAlAs,1064nm)five times for 3 weeks.	Local Corticosteroid Injection	——	Pain:VASDisability:-follow-up: 3 mouths
E. Kaydok et al (2020) [32]	Turkey	*n*=59age:44.2 ± 9.3Exp: 29（9m/20f)Con: 30（7m/23f) population：general population	HILT（YAG,1064nm):a total of nine treatment sessions over a period of three consecutive weeks	LILT(904nm):three times sessions per week were administered over a period of three weeks		Pain:VASDisability:QDASHfollow-up: -
A. Santamato et al (2009) [33]	Italy	*n*=70age:54.1 ± 9.0Exp: 35（15m/20f)Con: 35（13m/22f) population：general population	HILT(ASA,1064 nm):a total of 10treatment sessions over a period of 2consecutive weeks (5 days per week)	US:a total of 10treatment sessions over a period of 2 consecutive weeks (5 days per week)		Pain:VASDisability:QDASHfollow-up: -

Clinical heterogeneity

Among the 15 included randomised controlled trials, significant clinical heterogeneity existed in terms of study subjects and interventions. Analysed by tendinopathy location, 4 studies addressed rotator cuff tendinopathy [22, 25, 33, 35], 8 focused on lateral epicondylitis [21, 23, 24, 26–28, 30, 32], 1 focused on De Quervain tenosynovitis [31], 1 examined proximal hamstring tendinopathy [29], and 1 investigated Achilles tendinopathy [34]. These studies predominantly involved the general population, differing from many research efforts centred on specific athletic cohorts. Regarding gender distribution among study subjects, most studies recruited slightly more males than females, particularly in studies of lateral epicondylitis, consistent with the epidemiological characteristics of this condition.

Regarding intervention protocols, HILT parameters exhibited considerable variation. Laser wavelengths range from 808 nm to 1064 nm, with energy densities varying from 6 to 150 J/cm². Treatment frequency is 3 to 5 sessions per week, with a total course lasting 2 to 4 weeks. Control interventions also exhibited considerable variation: 1 study compared HILT with a placebo, 9 compared it with other interventions (such as US, ESWT, or bandaging), and 3 evaluated HILT as an adjunct to conventional physiotherapy. Notably, follow-up time points ranged from immediately after treatment completion to 16 weeks, potentially representing a key source of heterogeneity in assessing long-term efficacy.

Pain

Across 13 studies involving 629 patients, high-intensity laser therapy significantly improved pain compared with control groups (MD: -1.15; 95%CI: -1.73 to -0.58) (Fig. 3). Pain was assessed using VAS in 12 studies and NPRS in 1 study [29]. Data from both scales (0–10 points) were pooled for meta-analysis. No apparent systematic bias was observed in the contour plots. Although asymmetry was detected, missing values were distributed across both significant and non-significant regions (Fig. 5a). The study further conducted subgroup analyses based on the type of control intervention, including “HILT versus placebo”, “HILT versus other interventions”, and “HILT versus no intervention”. “Other interventions” include US, ESWT, LILT, kinesiologic taping, drug injection and bandage. In the placebo subgroup [24], HILT significantly reduced pain (MD: -1.00; 95%CI: -1.78 to -0.22); in other intervention subgroup, comprising 9 studies [21, 23, 26, 28, 30–33], moderate pain relief was demonstrated (MD: -0.78; 95%CI: -1.47 to -0.10); and in no intervention subgroup, comprising 3 studies [22, 25, 29], the most pronounced improvement in

Table 2 HILT characteristics of the included trials

Study	Device type	Wavelength (nm)	Mean output power (W)	Energy density (J/cm2)	Treatment time(min\s)	Peak power(W)	Total energy	Application protocols
Sai-Nan Zhanget al (2024) [21]	Light Force	810、980	11	ND	8	ND	ND	a single daily 8-minute session at 11.0 W, applied to the lateral epicondyle from 10–20 mm, over 8 consecutive days.
Morteza Beyrami et al (2023) [22]	GaAlAs	808	4	20	200s	8	800	12 sessions over 4 weeks (3 sessions/week), applying a Ga-Al-As laser (4W, 808nm, 20 J/cm²) for 20 s each to 10 tender points around the shoulder per session (total 200 seconds/session), using a 2.5 cm spacer.
Emine Esra Bilir et al (2024) [23]	ASA	ND	4	6	1.25	ND	ND	9 sessions over 3 weeks. The first 3 sessions applied laser (4W, 6 J/cm²) for 75 seconds in circular motion to the most painful area. The next 6 sessions applied laser (6W, 100-150 J/cm²) for 12.5 minutes in linear motion to the pain region.
			6	100-150	12.5	ND	ND	
Umit Dundar et al (2015) [24]	YAG	1064	10.5	0.36-1.78	15	3K	1275	a day for 15 days during a period of 3 weeks. Each session consisted of three phases applied to the lateral epicondyle, common extensor tendon, and extensor muscles: 1) fast manual scanning with fluencies of 510, 810, and 970 mJ/cm² (625 J total), 2) trigger point inactivation at the tendon with fluencies of 360, 510, and 610 mJ/cm² (25 J total), and 3) slow manual scanning with the same fluencies as phase 1 (625 J total), delivering a total of 1275 J per session.
A.M.Elsodany et al (2018) [25]	ASA	1064	10.5	20	15	3K	2050	3 sessions each week for 4 weeks with a total of 12 sessions. Each session involved fast scanning (20 J/cm², 1000 J), fixed point application (50 J total), and slow scanning (20 J/cm², 1000 J) over the rotator cuff and adjacent muscles, totaling 2050 J per session.

Table 2 (continued)

Study	Device type	Wavelength (nm)	Mean output power (W)	Energy density (J/cm2)	Treatment time(min\s)	Peak power(W)	Total energy	Application protocols
İbrahim Karacaet al (2022) [26]	BTL-6000	1064	4 6	6 12	1.25 12.5	12	ND	a total of 4 sessions for 2 week. Each session applied laser for 75 seconds at 4W, 6 J/cm² (analgesic effect) followed by 12.5 minutes at 6W, 12 J/cm² (biostimulation effect) to the most painful areas of the lateral epicondyle.
Angela Notarnicola et al(2014) [34]	YAG	810、980、1064	13	120	93s	ND	1200	10 daily sessions of combined cold air flow (−30°C) and HILT; 13 W power, 2.6 W/cm² power density, 93 s total application time per session) were applied to the Achilles tendon insertion site (≈10 cm² area). The total energy delivered was 1,200 J per session (120 J/cm²).
Sidika Buyukvural Sen et al (2024) [28]	YAG	1064	15	39	ND	ND	ND	15 consecutive sessions over three weeks. The most painful region of the lateral epicondyle using a Nd:YAG laser (1064 nm, 15 Hz, pulsed mode, average output 15 W, energy density 39 J/cm² per point).
Sachin Vermaet al (2022) [29]	Lite Cure	810、980	5	50	6	ND	1800	HILT was applied as monotherapy to the hamstring origin at the ischial tuberosity. The treatment area was 36 cm² (6x6 cm). Parameters were: wavelength 980/810 nm, average power 5 W, dosage 50 J/cm², total energy 1800 J per session, applied in continuous mode for 6 minutes.
Ali SALLI et al (2016) [27]	ND	ND	4 6	6 12	1.25 12.5	ND	ND	a total of 10 sessions over 2 weeks (5 days per week). Treatment was applied to the most painful area. The protocol involved 2 phases: during the first four sessions, laser was applied at 4W, 6 J/cm² (analgesic effect) for 75 seconds per session using a circular motion. In the subsequent 6 sessions, laser was applied at 6W, 100-150 J/cm² (biostimulation effect) for 12.5 minutes per session using a linear motion.

Table 2 (continued)

Study	Device type	Wavelength (nm)	Mean output power (W)	Energy density (J/cm2)	Treatment time(min\s)	Peak power(W)	Total energy	Application protocols
H. Yüksel et al (2025) [30]	Meetronic	ND	8	ND	3	ND	ND	3 times a week for 3 weeks. The lateral epicondyle area using a predefined epicondylitis protocol with 3 phases. Phase 1 and 2: 8W power was applied to a circular area (radius 1.0 cm) for 3 minutes each (analgesic effect). Phase 3: A dose of 80 J/cm² at 8W power was applied (anti-inflammatory effect). All patients also performed a standardized therapeutic exercise program.
			8	80	ND			
Z. Zaki et al (2025) [35]	Cube 4 plus med	610, 800, 905, 970	ND	24.78	5.27	20	2478	HILT was applied over the symptomatic shoulder area, specifically targeting the greater tubercle of the humerus and extending 5 cm in all directions, covering 100 cm². The protocol consisted of two phases: an initial phase of three sessions delivering a total energy of 2478 J (24.78 J/cm²) and a subsequent phase of seven sessions delivering 4678 J (46.78 J/cm²), administered three times per week.
				46.78	8.13		4678	
M. R. Nikoo et al (2024) [31]	GaAlAs	1064	4	50	5	ND	ND	5 sessions over three consecutive weeks (sessions spaced 3 days apart). The first dorsal compartment (site of De Quervain's tenosynovitis) using a GaAlAs diode laser (wavelength 1064 nm, output power 8W, average energy intensity 4 J/cm²). Laser was delivered in pulsed mode (frequency 50 Hz) via a scanning method. Each session lasted between five to seven minutes.

Table 2 (continued)

Study	Device type	Wavelength (nm)	Mean output power (W)	Energy density (J/cm2)	Treatment time(min\s)	Peak power(W)	Total energy	Application protocols
E.Kaydok et al (2020) [32]	YAG	1064	8	6	75s	ND	150	2 phases over a total of nine sessions (3 sessions per week for 3 weeks). Phase I (first 3 sessions) used an intermittent mode to provide analgesic effects, delivering 8 W for 75 seconds per session with an energy density of 6 J/cm², totaling 150 J per session. Phase II (remaining 6 sessions) used a continuous mode for biostimulation, delivering 6 W for 30 seconds per session with an energy density ranging from 120 to 150 J/cm². The laser was applied directly to the lateral epicondyle and surrounding painful areas.
			6	120-150	30s		ND	
A.Santamato et al (2009) [33]	ASA	1064	6	760	10	ND	150	Over 2 weeks (10 sessions, 5/week), each 10-minute session applied approximately 2050 J of total energy in 3phases to the shoulder region (including trapezius, deltoid, and pectoralis minor muscles, as well as trigger points). The protocol combined fast and slow manual scanning of muscular areas with stationary application on painful points, using fluencies between 510 to 710 mJ/cm².

ND indicates that the article does not provide a detailed account

pain was demonstrated (MD: -2.22; 95%CI: -2.88 to -1.56). Overall heterogeneity analysis revealed substantial heterogeneity between studies (I^2 = 89%; $p < 0.01$), though this varied across subgroups: The placebo subgroup comprised only one study and exhibited no heterogeneity. The other interventions subgroup demonstrated high heterogeneity (I^2 = 86%; $p < 0.01$), whilst the no intervention subgroup showed moderate heterogeneity (I^2 = 75%; $p < 0.05$).

Disability

Data from 503 patients across 10 studies indicated that HILT significantly improved disability compared with control groups, as measured by pooled mean differences (SMD: -1.00; 95%CI: -1.77 to -0.22, $p < 0.05$) (Fig. 4). Despite the asymmetry, we detected no discernible publication bias through contour funnel plots (Fig. 5b), with both significant and non-significant data lacking. Subgroup analyses by disease type revealed non-statistically significant functional improvement in the elbow-related disorders subgroup comprised 7 studies [23, 24, 26–28, 30, 32] (SMD: -0.28; 95%CI: -0.59 to 0.03, p = 0.07); whereas the shoulder-related disorders (Shoulder) subgroup, comprising 3 studies [22, 25, 33], demonstrated a larger effect size yet similarly failed to achieve statistical significance (SMD: -3.43; 95%CI: -7.72 to 0.86, p = 0.12). Analysis of overall heterogeneity across all included studies revealed substantial inter-study heterogeneity (I^2 = 94%; $p < 0.01$), with the shoulder-related disorders subgroup exhibiting extremely high heterogeneity (I^2 = 98%, $p < 0.01$), whilst the elbow-related disorders subgroup demonstrated moderately high heterogeneity (I^2 = 50%, p = 0.06).

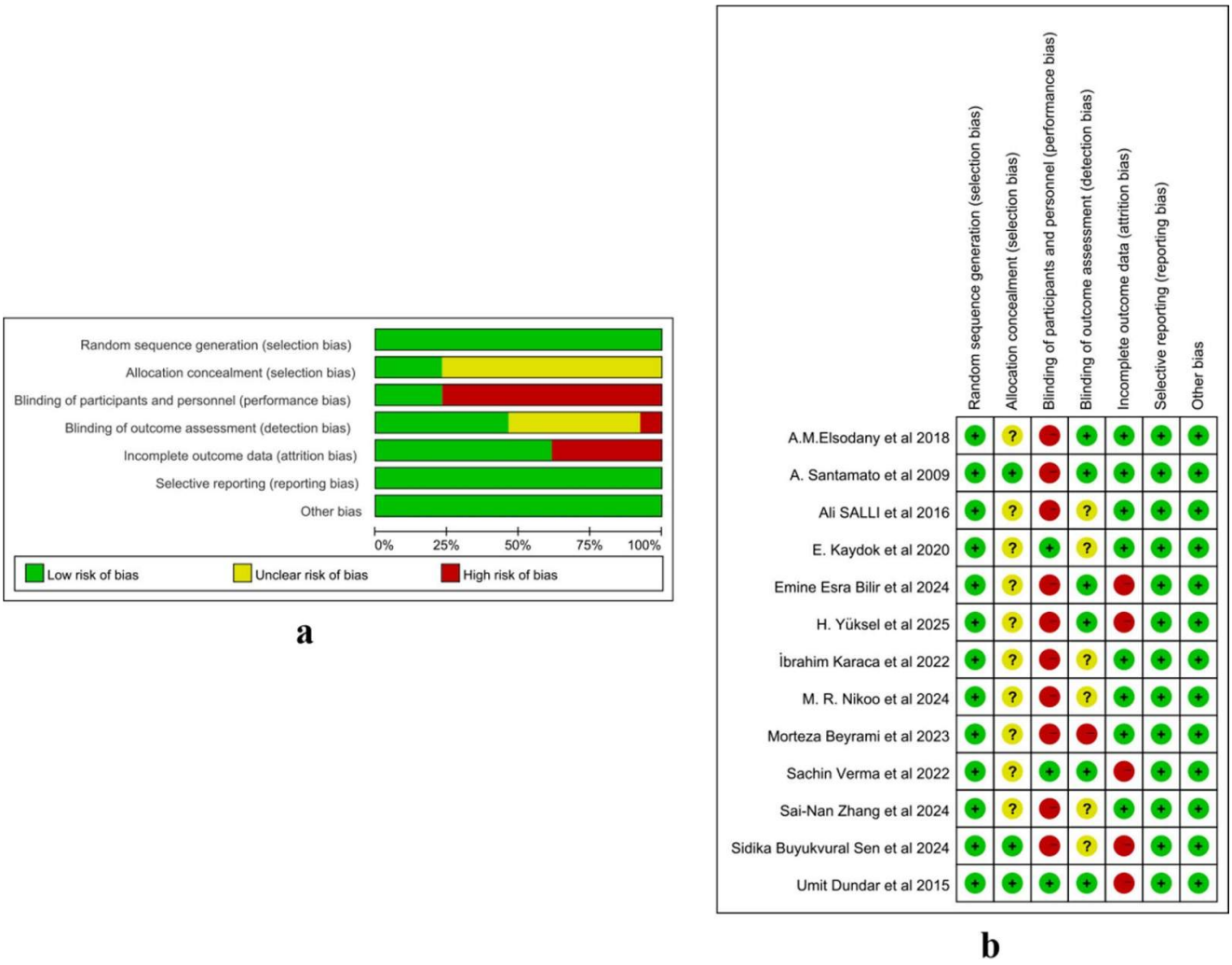


Fig. 2 Risk of bias plot. **a** Risk of bias plot; **b** Summary of risk of bias

VAS subgroup analysis by pain status

In an analysis encompassing 4 studies and 179 patients that assessed pain using VAS, HILT significantly improved pain symptoms across different states (MD: -0.95; 95%CI: -1.54 to -0.36) (Fig. 6). Subgroup analyses were conducted according to pain status, including resting pain, movement pain, nocturnal pain, and stress-induced pain. In the resting pain subgroup (4 studies [23, 24, 26, 28], 179 patients), HILT showed near-significant improvement (MD: -0.84; 95%CI: -1.71 to 0.04, p = 0.06). In the exercise pain subgroup (3 studies [23, 26, 28], 118 patients), similarly near-significant improvement was observed (MD: -0.86; 95%CI: -1.74 to 0.02, p = 0.06) the nocturnal pain subgroup (1 study [23], 40 patients) showed no significant difference (MD: 0.00; 95%CI: -2.09 to 2.09) whereas the stress-related pain subgroup, comprising one study with 61 patients [24], demonstrated the most pronounced pain reduction (MD: -2.30; 95%CI: -3.39 to -1.21, p < 0.01). Comparisons between subgroups of different pain states revealed significant heterogeneity (I^2 = 52.6%, p = 0.10), indicating that HILT may exhibit differential therapeutic effects across pain types. Notably, HILT produced the most pronounced improvement for stress-related pain, while exerting limited influence on nocturnal pain.

Subgroup analysis by follow-up period

Subgroup analysis conducted according to follow-up period (Table 3) revealed a clear association between follow-up duration and treatment efficacy. Regarding pain assessment, no significant improvement was observed during short-term follow-up (≤8 weeks) (MD: 0.22, 95%CI: -0.81 to 1.26, p=0.67), while significant improvement was observed at mid-term follow-up (9–16 weeks) (MD: -1.50, 95%CI: -2.34 to -0.66, p<0.01). The greatest improvement was noted at long-term follow-up (>16 weeks) (MD: -3.03; 95%CI: -3.36 to -2.70, p<0.01). Regarding disability assessment, no

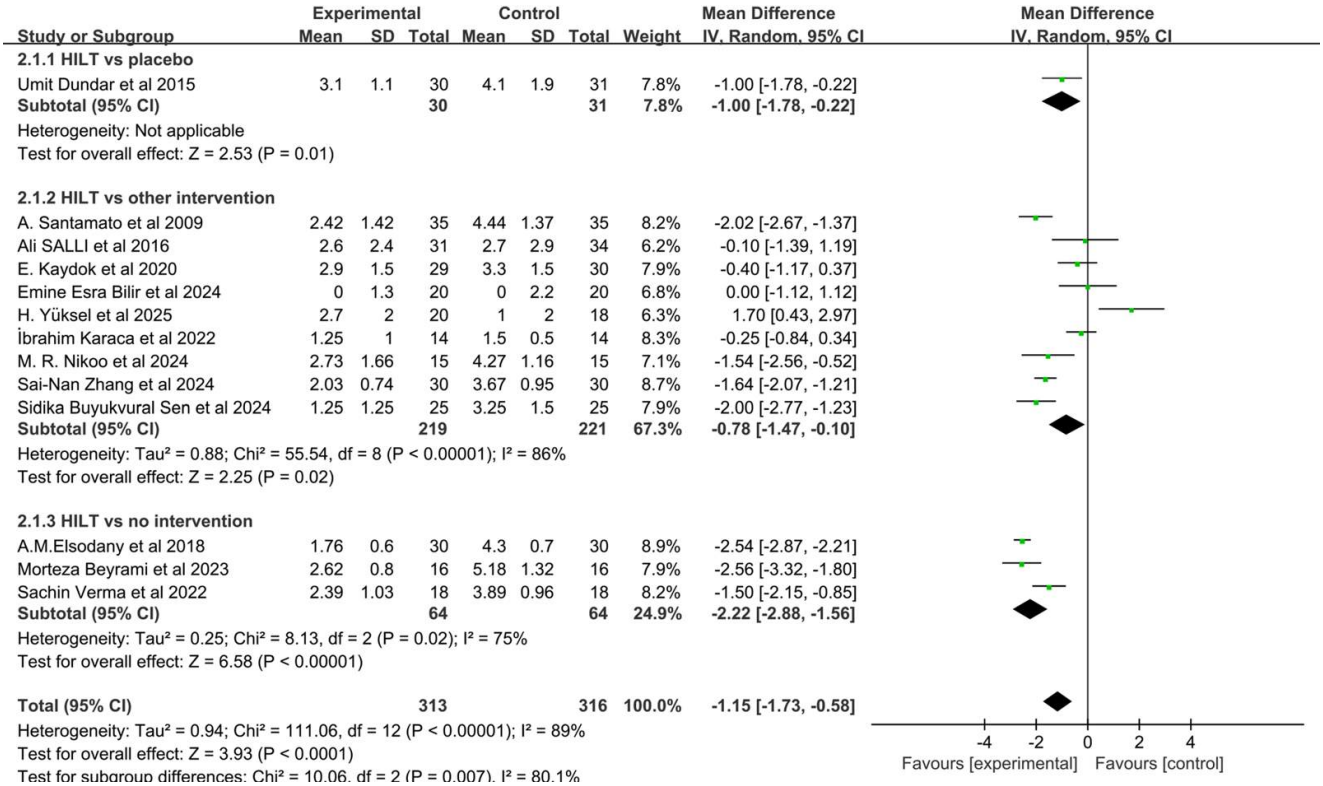


Fig. 3 Random-effects forest plot for pain

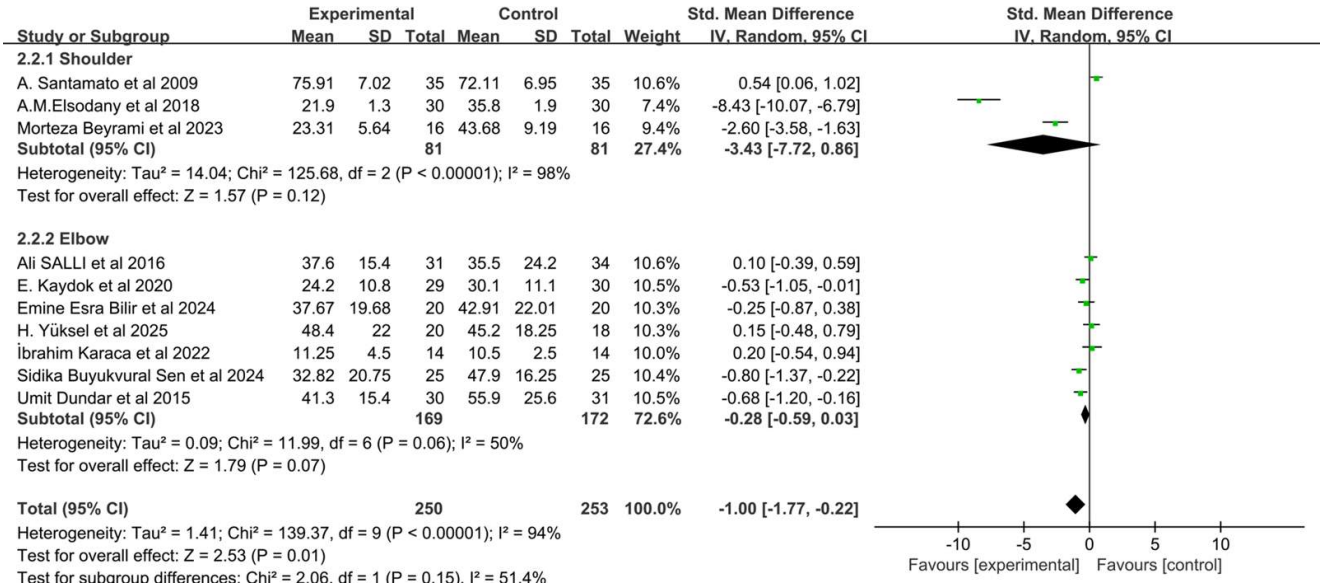


Fig. 4 Random effects forest plot for disability

significant improvement was observed at short-term follow-up (≤8 weeks) (SMD: -0.73, 95%CI: -1.67 to 0.20, $p=0.13$). Significant improvement was noted at mid-term follow-up (9–16 weeks) (SMD: -3.24, 95%CI: -5.92 to -0.56, $p<0.05$), while long-term follow-up (>16 weeks) demonstrated the greatest improvement (SMD: -9.46, 95%CI: -11.28 to -7.63, $p<0.01$). Heterogeneity analysis revealed moderate to high heterogeneity across all follow-up subgroups, particularly pronounced in the mid-term follow-up subgroup (pain I^2 = 94%, disability I^2 = 97%; both $p<0.01$). This indicates

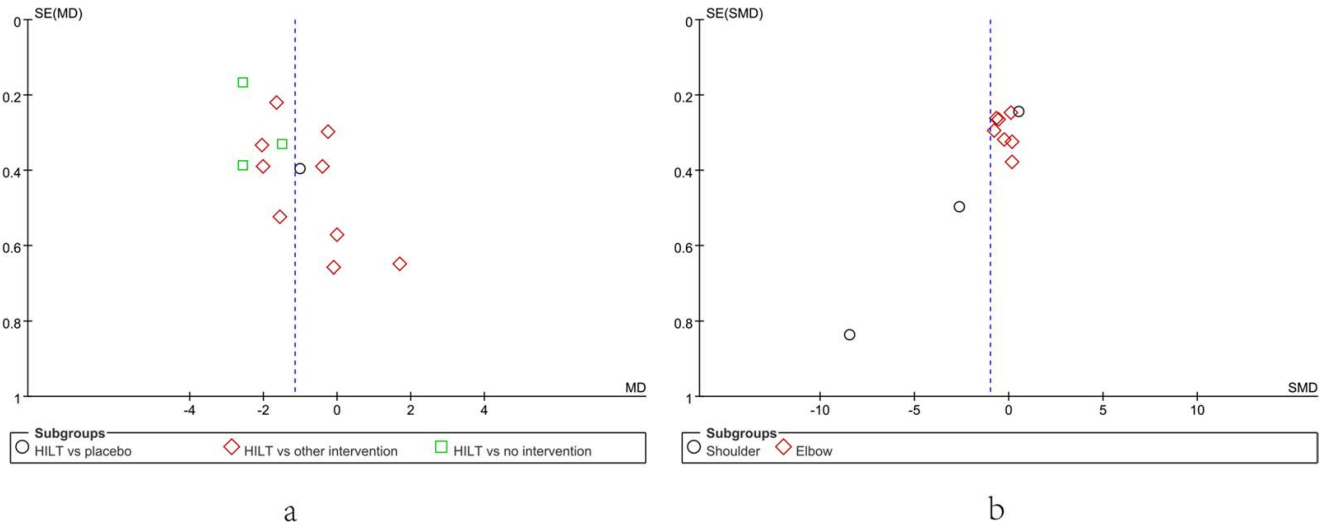


Fig. 5 Funnel diagram. **a** Pain; **b** Disability

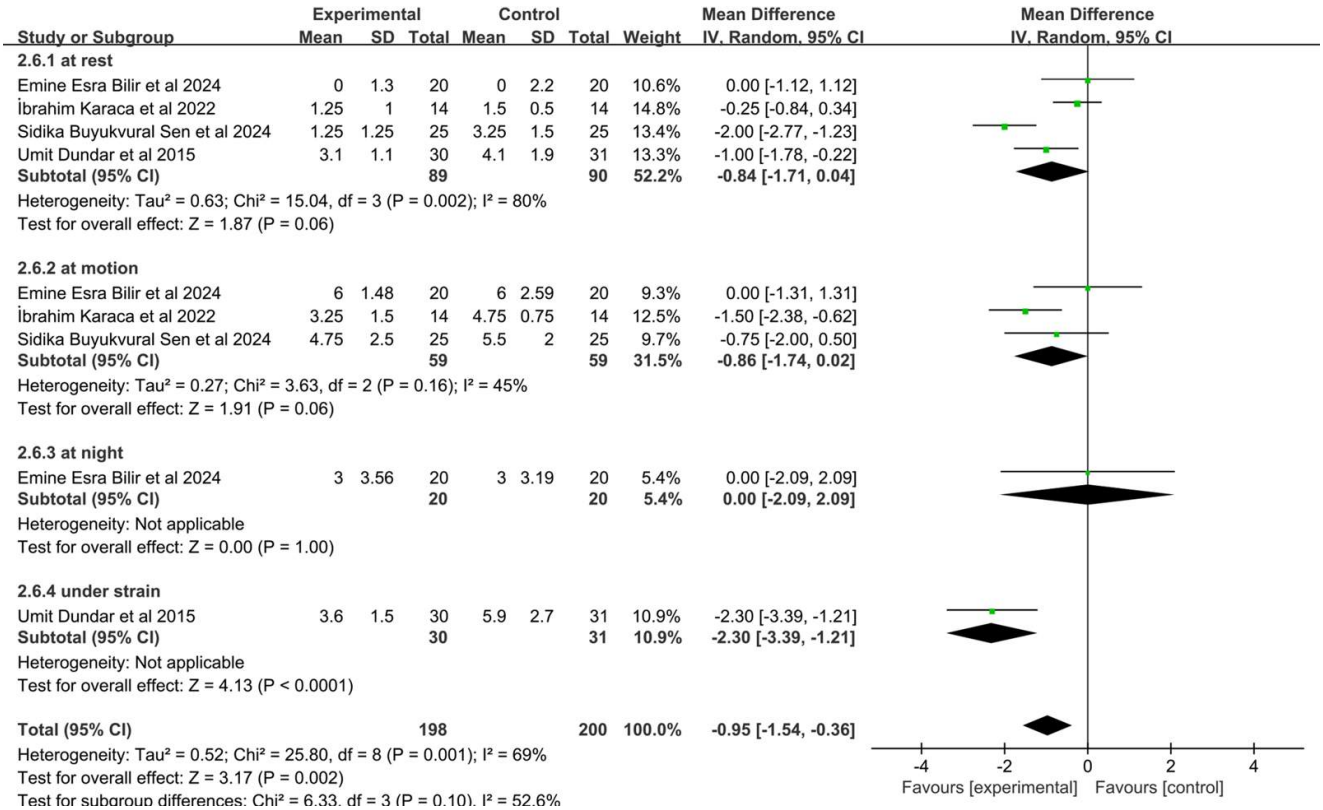


Fig. 6 Forest plot of subgroup analysis by VAS status

that the analgesic and disability recovery effects of HILT increase over time, with significant benefits observed after the mid-term period.

Descriptive analysis

Notarnicola et al. [34] conducted a randomised controlled trial involving 60 patients with chronic insertional Achilles tendinopathy, following exclusion of quantitative studies. Employing a distinct intervention, it compared the efficacy of cold air combined with high-energy laser therapy (CHELT) against ESWT. Outcomes were assessed using VAS, Ankle–Hindfoot Scale, and Roles and Maudsley Score. Results demonstrated that at the conclusion of treatment and at 2-month and 6-month follow-ups, the CHELT group exhibited significantly lower VAS scores (T3: 1.7 ±

1.0 vs. 3.3 ± 2.4; $p < 0.05$), ankle-hindfoot function (T2: 81.1 ± 9.2 vs. 73 ± 22.9; $p < 0.05$), and patient satisfaction (T3: 1.5 ± 0.5 vs. 1.5 ± 0.5; $p < 0.01$). The study reported no significant adverse reactions.

Zaki et al. [35] enrolled 46 patients with ultrasound-confirmed partial-thickness tears, randomly assigned to either the HILT group ($n = 23$) or the sham laser control group ($n = 23$). Outcomes comprised VAS and CMS. Results indicated that at the conclusion of treatment, the HPLT group demonstrated a mean reduction in VAS of 3.2 points, compared with a reduction of 1.4 points in the control group. CMS scores improved by 24.1 and 11.7 points, respectively, with statistically significant intergroup differences ($P < 0.01$). However, due to a lack of post-intervention means and standard deviations, the results could only be included as a qualitative description.

Safety analysis

Among the 15 studies included, the majority explicitly reported no serious adverse events associated with HILT treatment. There were no instances of treatment discontinuation or study withdrawal due to adverse reactions.

Discussion

This systematic review and meta-analysis synthesised results from 15 randomised controlled trials evaluating the efficacy and safety of HILT for various tendinopathies. Findings indicate that compared with control interventions, HILT demonstrated significant advantages in both pain relief (MD: -1.15; 95%CI: -1.73 to -0.58 and disability improvement (SMD: -1.00; 95%CI: -1.77 to -0.22, $p < 0.05$). Subgroup analyses further revealed differential HILT efficacy across tendon locations: The therapeutic efficacy for shoulder-related disorders (MD: -2.54) was superior to that for elbow-related disorders (MD: -0.91). Moreover, subgroup analysis by follow-up duration indicated that HILT's therapeutic effects accumulated over time, with the most pronounced improvement observed in long-term follow-ups (> 16 weeks). Subgroup analysis by VAS status revealed that HILT's pain-relieving effect under stress (SMD: -2.30) was most pronounced, providing crucial evidence for individualised clinical treatment.

Mechanisms of HILT in pain management

Regarding the mechanism of action underlying HILT's analgesic effects, existing research offers multifaceted explanations. Firstly, HILT exerts analgesia through direct neural conduction blockade; when laser irradiation intensity reaches 270 mW/cm² or higher, it rapidly interrupts pain signal transmission and inhibits depolarisation of C-fibre afferent neurons [40, 41]. Secondly, according to gate control theory, HILT activates large-diameter Aβ fibres, stimulating inhibitory interneurons in the superficial layers of the dorsal horn of the spinal cord. This subsequently inhibits the upward transmission of nociceptive signals from C and Aδ fibres [42]. It is noteworthy that although pain and thermal sensation share Aδ and C fibre afferents [43], the gate control mechanism may preferentially inhibit the nociceptive signal pathway, thereby preserving normal temperature perception. At the cellular and molecular level, HILT's analgesic mechanism is closely linked to improved mitochondrial function. Cytochrome c oxidase, acting as the primary photoreceptor, absorbs photons to accelerate electron transport reactions. Concurrently, HILT induces photodissociation of nitric oxide from its binding site, restoring enzyme activity, increasing ATP production, and reducing oxidative stress [44, 45]. Furthermore, laser therapy activates endogenous pain regulation systems, including increased β-endorphin release, enhanced serotonin secretion, reduced bradykinin levels, and modulation of neuronal excitability through normalisation of Ca^{2+}, Na^{+}, and K^{+} concentrations [46–48]. HILT further improves local microcirculation by stimulating vasodilation, thereby facilitating nutrient transport and metabolic waste clearance while effectively reducing pro-inflammatory cytokine levels [49, 50]. The synergistic interaction of these multiple mechanisms confers unique advantages to HILT in pain management, providing a robust theoretical foundation for tendinopathy treatment. Moreover, the therapeutic response to HILT may be influenced by the specific stage of tendon pathology and the clinical subtype of tendinopathy. Different pathological phases may require tailored HILT parameters to optimize photobiomodulation and photothermal effects, though current evidence remains limited in this regard.

Efficacy on pain and comparison

Compared with the control group, this study found that HILT demonstrated a significant advantage in alleviating pain (MD: -1.15; 95%CI: -1.73 to -0.58). Importantly, this effect exceeds the 1.0-point minimum clinically important difference (MCID) for VAS, confirming its clinical relevance [51]. However, it should be noted that the confidence intervals for some pooled estimates crossed the MCID threshold, suggesting that the clinical benefit may not be consistent across all patient populations or treatment settings. This finding is consistent with the results of Akkurt et al. [52] in their study on lateral epicondylitis of the humerus; they observed that HILT proved more effective than conventional treatment methods in alleviating chronic

pain. Furthermore, a key innovation of this study lies in its first systematic analysis of HILT's differential efficacy across distinct pain states, revealing its most pronounced analgesic effect on stress-induced pain (MD: -2.30; 95%CI: -3.39 to -1.21). This aligns with findings by Dundar et al. [24], who observed particularly marked improvements in stress-related pain among patients with lateral epicondylitis following HILT treatment. This phenomenon may relate to increased tissue oxygen demand, metabolic by-product accumulation, and heightened mechanical stimulation during stress states, where HILT's photothermal effects and blood flow enhancement prove more pronounced. From a clinical perspective, the marked improvement in stress-induced pain observed with HILT suggests it may be particularly well-suited for patient groups requiring a return to sport or restoration of functional activities.

Efficacy on disability and site-specific effects

In terms of disability function improvement, this study demonstrated that HILT produced a moderate and statistically significant overall effect on disability (SMD: -1.00; 95%CI: -1.77 to -0.22, $p < 0.05$). This functional benefit may be attributed to mechanisms beyond pain relief: HILT is proposed to directly enhance tendon repair by promoting collagen synthesis and restoring extracellular matrix integrity, thereby improving the tendon's capacity to withstand functional loads. Concurrently, its analgesic effect facilitates earlier and more effective engagement in rehabilitation, which is crucial for restoring task-specific movements. This mechanistic basis supports the observed differential therapeutic efficacy across different tendon locations [53]. Our findings align with the perspective of Stasinopoulos et al. [54], who posited that treatment outcomes are closely linked to the anatomical position of the tendon, its blood supply, load characteristics, and pathological stage. This study observed superior efficacy of HILT in treating rotator cuff tendinopathy compared to other sites, potentially attributable to the shoulder region's rich vascular network, which facilitates laser-induced photothermal effects and biostimulation. The substantial pain reduction in the shoulder subgroup (MD: -2.54) far surpasses the typical MCID for shoulder disorders (~ 1.4 points) [55]. This finding is corroborated by related studies [10, 25], which demonstrated HILT's most pronounced therapeutic effect in shoulder pain conditions such as rotator cuff tendinopathy, likely due to its targeted biological response in specific tissues.

Cumulative time effect of HILT

Regarding the time-dependent nature of therapeutic effects, data analysis indicates that HILT efficacy increases with prolonged follow-up duration, with the greatest improvement observed in the long-term follow-up group (> 9 weeks). This finding aligns with Bjordal et al.'s [56] theory on the cumulative effects of laser therapy, which posits that laser irradiation promotes progressive therapeutic outcomes through sustained cellular metabolic regulation and tissue repair. Indeed, Notarnicola et al. [57] similarly observed in Achilles tendinopathy studies that HILT treatment yielded superior outcomes at 6-month follow-up compared to 2 months, suggesting enhanced efficacy over time. A systematic review further indicates that significant tissue remodelling from biomechanical and histological improvements in tendon healing following laser therapy typically requires 4 to 8 weeks to manifest [58].

Clinical and practical implications

To extrapolate the disability improvement outcomes of HILT to clinical practice requires consideration of its advantages relative to other treatment approaches. Compared to low-level laser therapy (Power is typically < 500 mW), HILT demonstrates significant advantages in penetration depth (Power typically exceeds 1 W), mechanism of action, and long-term efficacy. Relevant studies confirm that HILT achieves deep tissue penetration of 5–6 cm, far exceeding the typical limitation of low-level lasers, which rarely exceed 2 cm [59]. However, the high scattering properties of biological tissue significantly limit its effective therapeutic depth [60]. Consequently, the practical efficacy of HILT for deep tendons may stem more from the indirect systemic or segmental physiological effects it induces, rather than direct local action on the pathological site. Consequently, HILT is more suitable for treating deep tendon pathologies. More importantly, HILT combines non-thermal photobiomodulation with controllable photothermal effects, enabling synergistic analgesia, anti-inflammation, and tissue repair within the same treatment cycle. This provides a multi-targeted, integrated solution for chronic tendinopathies. Multiple comparative studies against conventional interventions indicate that HILT outperforms extracorporeal shock wave therapy in pain reduction and functional improvement while generating fewer adverse reactions. This conclusion is corroborated across numerous studies on tendinopathy treatment [27, 57, 61]. While HILT may not yield as pronounced short-term therapeutic effects as corticosteroid injections, relevant studies reveal its long-term therapeutic effects are more enduring [33]. Crucially, it avoids severe treatment risks such as tissue atrophy or tendon rupture. This safety advantage renders HILT particularly suitable for chronic tendinopathy patients requiring long-term therapeutic management. Owing to its deep penetration, multifaceted action, and favourable safety profile, HILT is now regarded as an ideal choice for the long-term management of deep or chronic tendinopathies.

Limitations and future directions

However, several limitations should be considered. Firstly, most included studies had small sample sizes and short follow-up periods (≤ 16 weeks), limiting assessment of long-term efficacy. Secondly, considerable heterogeneity existed in HILT parameters and control interventions across studies, which may reflect inadequate consideration of tendinopathy stages (e.g., reactive vs. degenerative) as described in Cook's continuum model [62]. Thirdly, methodological weaknesses such as inadequate blinding and allocation concealment were common, potentially introducing bias. Additionally, combining outcomes measured with different tools may have increased heterogeneity. Although subgroup analysis could address this, limited studies and data precluded a thorough analysis. Future large-scale, multicenter RCTs with longer follow-up are needed to optimize HILT parameters, standardize protocols, and evaluate efficacy across patient subtypes.

Conclusions

Overall, HILT represents a safe and effective treatment for tendinopathy, demonstrating significant improvements in pain and functional impairment. It exhibits superior efficacy for rotator cuff tendinopathy compared to other sites, provides optimal relief for stress-induced pain, and shows enhanced therapeutic effects with prolonged follow-up. However, the studies included in this review exhibit considerable heterogeneity. Future research of higher quality and larger sample sizes is required to determine optimal treatment parameters and protocols, evaluate long-term efficacy, and explore optimised combination strategies with other therapeutic approaches.

Abbreviations

HILT	High-intensity laser therapy
RCTs	Randomized controlled trials
RoB	Collaboration's Risk of Bias
VAS	Visual Analogue Scale
NPRS	Numerical Pain Rating Scale
SPADI	Shoulder Pain and Disability Index
PRTEE	Patient-Related Tennis Elbow Evaluation
DASH	Disability of the Arm and Shoulder
PEDro	Physiotherapy Evidence Database
CMS	Constant Murley Scale
LE	Lateral epicondylitis
RCT	Rotator Cuff Tendinitis
AT	Achilles Tendinosis
PHT	Proximal Hamstring Tendinopathy
CHELT	Cold air combined with high-energy laser therapy
ESWT	Extracorporeal Shock Wave Therapy
US	Ultrasound
AT	Achilles tendinopathy
LILT	Low-Intensity Laser Therapy
MCID	Minimum Clinically Important Difference

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10103-026-04842-3.

Acknowledgements Not applicable.

Author contributions All authors contributed to the article and agreed to submit the manuscript. Rui Hong: Conceptualization, data curation, methodology, software, visualization, writing original drafts, and writing review editing. Xiang-Ming Lin: Data curation, formal analysis, writing original draft, and writing review editing. Yi-Shan Xue and Yu-Han Liu: Data curation, formal analysis, software, writing original draft, and writing review editing. Wan-Rong Xu and Tian-Cheng Li: Formal analysis, software, and supervision. Ying Li : Methodology, and supervision. Ben-Guo Wang: Conceptualization, and supervision.

Funding This work was supported by the Shenzhen Longgang District Science and Technology Innovation Bureau (LGKCYLWS2023004) and the Shenzhen Basic Research Project of Natural Science Foundation (JCYJ20250604180120026).

Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Consent to participate 1. Informed consent was obtained from all individual participants included in the study.
2. Informed consent was obtained from legal guardians.
3. Written informed consent was obtained from the parents.
4. Verbal informed consent was obtained prior to the interview.

Clinical trial number No application.

Competing interests The authors declare no competing interests.

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